

Multiphase Melt Inclusions in the Jinchuan Complex, China: Implications for Petrogenic and Metallogenic Physico-Chemical Conditions

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Abstract

Three types of melt inclusions were identified in pyroxenes, olivine, and chromite from the Jinchuan intrusion—sulfide inclusions (I), crystallized polyphase inclusions (II), and fluid inclusions (III). Sulfide inclusions in chromite are spherical and occur parallel to chromite growth planes. Polyphase inclusions are found in olivine and consist of olivine, orthopyroxene, chrome-spinel, amphibole, and sulfide daughter minerals. Primary fluid inclusions are usually found in olivine and pyroxene with spherical to tubular shapes.

Laser Raman analyses were used to identify daughter minerals and fluid phases in the inclusions. The result shows that fluids consist mainly of CO₂, with minor amounts of SO₂, H₂S, and CH₄. This indicates that the parental magma of the Jinchuan intrusion contains abundant volatiles.

Microthermometry shows that Type-II inclusions in dunite and sulfide-rich lherzolite have higher trapping temperatures (1000° to 1050°C), whereas trapping temperatures in sulfide-poor lherzolite are lower (900° to 950°C). The result of low-temperature microthermometry indicates that the CO₂ homogenization temperatures of +10° to +15°C correspond to depths of 12.9 to 15 km (deep-level magma chamber) and the homogenization temperatures of +26° to +29°C correspond to depths of 6.9 to 9.3 km (upper magma chamber). The immiscible sulfide in the inclusions demonstrates that segregation of the sulfide melt occurs at an early stage during magmatic differentiation.

Introduction

THE JINCHUAN DEPOSIT of eastern China, a major source of Ni, Cu, and platinum-group elements (PGE), was discovered in 1958. Since then, the Sixth Geological Team (SGT) of the Gansu Geological Survey, China has worked on this ore-bearing intrusion. Little information has been published during the past 30 years, although recently some studies have appeared that have addressed the general geology, sulfide mineralogy (SGT, 1984; Gang et al., 1993), and REE and PGE chem-

istry (Sun, 1986; Yang, 1989, 1994; Yang et al., 1991, 1992; Gang and Naldrett, 1992).

In past studies of the intrusion, a number of issues remained unresolved, especially aspects of petrogenic and metallogenic physico-chemical conditions. The mechanism by which the sulfide melt separated from the primary magma also is unclear. The significance of melt inclusions as a tool for understanding these geological processes has increased substantially during the past two decades. For ore environments such inclusions provide valuable information on the

temperature and pressure conditions attending ore-forming processes (Roedder, 1979). The most efficient way to address such questions about processes is to study the primary melt inclusions trapped in minerals.

This paper focuses in particular on the types and occurrences of melt inclusions identified in the Jinchuan complex, and then discusses the segregation and precipitation sequence of the sulfides. The purpose is to explore the mechanisms leading to sulfide immiscibility and to present new information on the genesis of sulfide ore deposits.

Geology of the Jinchuan Intrusion

Cu-Ni sulfide deposits of magmatic origin are known to exist in three different tectonic settings—in the interiors of stable cratonic blocks, in active continental margins, and in forearc and interarc troughs. However, all of the largest deposits—including Bushveld, Stillwater, and Sudbury—originated in active continental-margin tectonic settings. The Jinchuan deposit also formed in such an active continental zone.

The Jinchuan intrusion is located at the margin of the Sino-Korean massif (Fig. 1). It intruded into late Archean metasedimentary rocks consisting of various gneisses, chlorite-quartz schists, banded marbles, and migmatite (Tang, 1991).

The Jinchuan intrusion is a NW-oriented lens, over 6000 m long, ~200 to 300 m wide, and >1100 m deep (SGT, 1984), exhibiting a crude zonation of units around the central, Mg-rich ultramafic body. A sub-marginal zone of websterite and plagioclase websterite, ~50 m thick, is consistently developed between the lherzolite cumulates and the marginal schist. The marginal zone, 1 to 4 m thick, occurs as the “outer crust” of the intrusion, and consists of chlorite-tremolite schist. Gravity data, contact relations, and internal layering suggest that the intrusion has the form and structure of a basin (Yang, 1994).

The intrusion is divided into four parts by E-W to NE-SW translational faults; these parts are referred to as Mines III, I, II, and IV, from west to east. The bulk of the intrusion (including Mines I and II) forms outcrops, although this is not the case for Mines III and IV (Yang, 1994).

According to petrographic studies, the intrusion consists of three main rock types. In decreasing order of abundance, they are: lherzolite, dunite, and websterite. The principal mineral assemblage of this intrusion is olivine + pyroxene + plagioclase. Olivine and pyroxene are commonly fresher than plagioclase. Accessory minerals are chromite, amphibole, apatite, and biotite. The primary igneous mineralogy is well preserved in the south central part of the intrusion (Mines II and IV), and the degree of alteration increases to the north (Mines I and III).

The primary igneous minerals apparently crystallized in the following order: chromite, olivine, orthopyroxene, clinopyroxene, and plagioclase. Chromite occurs mainly as an accessory mineral and occasionally is enclosed in olivine. Olivine occurs as medium-grained euhedral to subhedral crystals. Most of the olivine is chrysolite (Fo_{90-77}), and minor amounts are forsterite (Fo_{91-94}) (SGT, 1984). Orthopyroxene (En_{88-72}) is relatively less abundant than clinopyroxene, which crystallized as subhedral crystals and consists of endiopside ($\text{En}_{45-62} \text{Wo}_{37-45} \text{Fs}_{1-11}$) and augite ($\text{En}_{40.5-43.5} \text{Wo}_{38.5-45} \text{Fs}_{14.5-18.0}$). Plagioclase (average An_{63}) is present as allotypic granular crystals and forms a poikilitic texture with olivine.

The sulfide deposit consists of hundreds of orebodies. Most of them occur in the middle and bottom parts of the intrusion. The volume of metallic sulfide minerals generally increases toward the base of the complex, where the mineral assemblage is pyrrhotite, pentlandite, and chalcopyrite. Sulfides form medium- to fine-grained interstitial aggregates between olivine and pyroxene. The dominant texture is that of a mesocumulate (Fig. 2A).

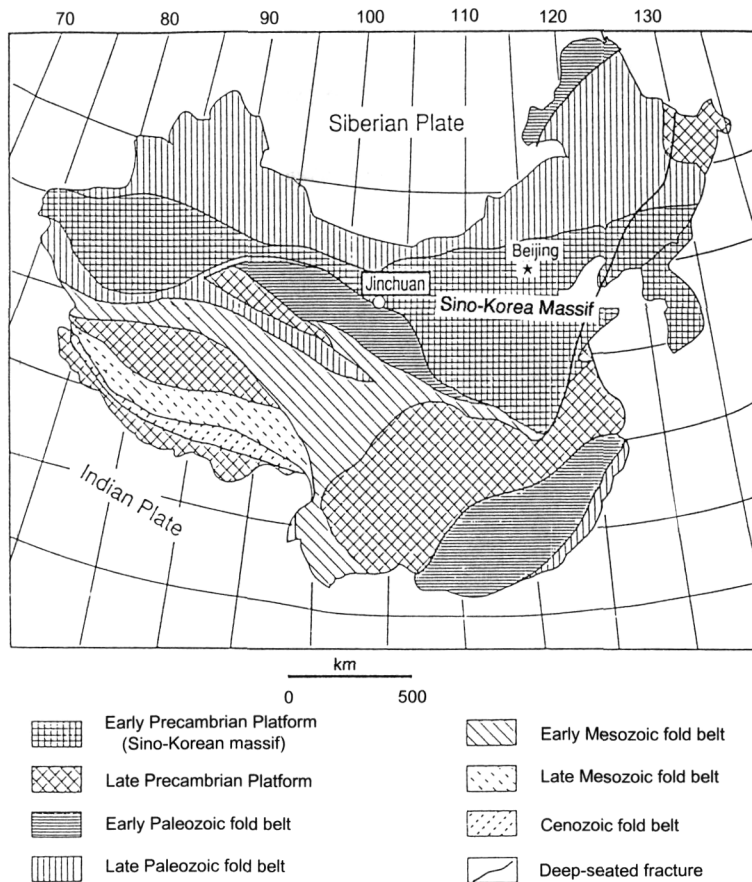


FIG. 1. Geological map showing the location of the Jinchuan region in relation to the main tectonic units of China (after Gang and Naldrett, 1992).

Lherzolite

Lherzolite can be divided into sulfide-rich and sulfide-poor phases (Yang, 1994). Most of the sulfide-rich lherzolite has been altered, and fresh olivine and pyroxene are recognized only partially. Pyroxenes are frequently replaced by amphibole and chlorite. Under these conditions, sulfide grains are found in association with hydrous phases such as biotite, apatite, and amphibole.

Sulfide-poor lherzolite contains fresh olivine and pyroxenes. Olivine is generally less serpentinized, and fresh olivine grains are common; olivine is poikilitically enclosed by pyroxene. The pyroxenes (characteristically interstitial to cumulus olivine) may occur as

reaction rims around olivines or as oikocrysts enclosing cumulus olivine grains.

Dunite

Dunite constitutes about 30% of the intrusion—usually occurring at the bottom of the intrusion—and is almost entirely sulfide bearing. It consists of 90 to 95 mod% olivine, 1 to 6 mod% pyroxene, and minor plagioclase and chromite, as well as interstitial sulfide minerals. Olivine occurs as medium-grained euhedral to subhedral crystals and is intensively altered to serpentine and magnetite. In extreme cases, all of the olivine is serpentinized, and only the outlines of its grain shapes remain discernible. Various defor-

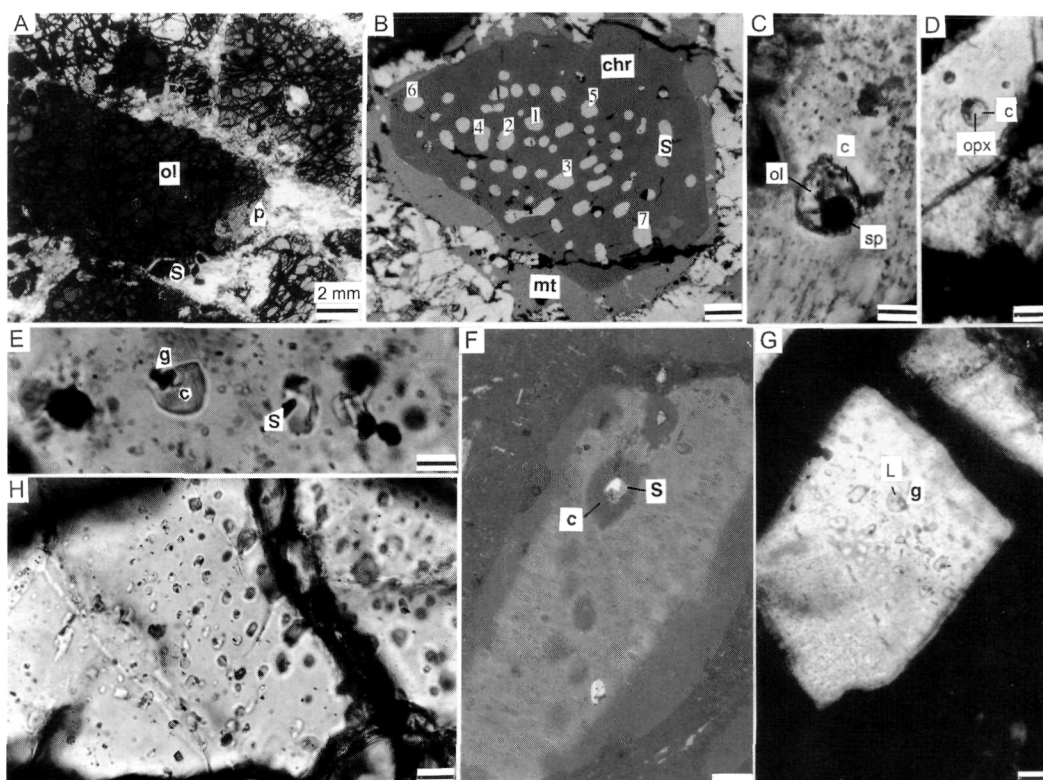


FIG. 2. A. Photomicrograph of lherzolite with an olivine (ol) + pyroxene (p) + sulfide (s) cumulate texture. B–H. Photomicrographs of solid and fluid inclusions in olivine. Scale bars are 30 μm . Abbreviations: ol = olivine; mt = magnetite; sp = spinel; opx = orthopyroxene; g = vapor bubble; c = unidentified; L = liquid; s = sulfide. B. Type-I sulfide inclusions in chromite (chr), showing that the occurrence of primary inclusions is parallel to growth surfaces within chromite. C. Type-II crystallized polyphase inclusions with spinel (sp), ol, and unidentified (c) daughter minerals. D. Type-II crystallized polyphase inclusions with opx and c; the negative nature of the inclusions reflects the crystal structure of the host olivine; an orthopyroxene daughter mineral is located in the central area of the inclusions. E. Type-II inclusions with sulfide daughter minerals and vapor bubble in the inclusion. F. Type-I sulfide inclusions in olivine. G. Type-III primary fluid inclusions, in which the larger inclusions have nucleated vapor bubbles and do not reach the surface of the host mineral. H. Type-III secondary fluid inclusions, in which the larger inclusions have nucleated vapor bubbles. Figure 2F obtained under transmitted plus reflected light; Figures 2A–2E and 2G–2H were obtained under transmitted light.

mation textures are found in some dunite and sulfide-rich lherzolite (SGT, 1984).

Analytical Techniques

Doubly polished thin sections were studied by JEOL-733 electron microprobe analyses. The detector conditions were 15 KV accelerating voltage, 10 nA specimen current, and a beam diameter of $\sim 5 \mu\text{m}$. The

following standard materials were used: synthetic MgO , Al_2O_3 , TiO_2 , Cr_2O_3 , MnO , and NiO for $\text{MgK}\alpha$, $\text{AlK}\alpha$, $\text{TiK}\alpha$, $\text{CrK}\alpha$, $\text{MnK}\alpha$, and $\text{NiK}\alpha$, respectively; and natural hematite, adularia, wollastonite, tugtupite, apatite, and albite for $\text{FeK}\alpha$, $\text{K}\alpha$, $\text{CaK}\alpha$, $\text{ClK}\alpha$, $\text{NaK}\alpha$, and $\text{SiK}\alpha$, respectively. The accelerating voltage was 20 to 25 kv at a reference beam current of 20 nA, with a beam diameter of $5 \mu\text{m}$ for sulfides. The standards used were

chalcopyrite for CuK α , FeK α , and SK α ; the synthetic metallic Ni for NiK α ; and Co for CoK α . Data were reduced using a computer program prepared by JEOL (Hirata, 1985). The program includes atomic number corrections by Philibert and Tixer (1968), an absorption correction by Philibert (1963), and characteristic fluorescence corrections by Reed (1965).

The Laser Raman Microprobe was used to study the inclusions. All determinations were conducted at room temperature (20°C) on a Laser RAMANOR U-1000 microspectrometer at the Xi'an Research Institute of Geology and Mineral Resources, China. A 400 mW Ar⁺ laser was used to emit a monochromatic beam with a wavelength of 514.5 nm passing through 400 to 500 μ m slits. The relative effective cross sections of Raman scattering were used to calculate the mole fraction of each component according to the formula of Schrotter and Klockner (1979).

The inclusions in olivine were studied using a Leitz 1350 high-temperature microthermometer, according to Roedder's (1984) method of temperature measurement. Prior to each run, the sample chamber was flushed with dry nitrogen gas at ambient temperature for at least 25 minutes. Calibration was performed using Au (melting point 1064°C; West, 1984). During runs, the samples were brought to temperatures between 800° and 1100°C at 20°C per minute, and then held at this temperature for 10 to 20 minutes. Each heating step was followed by several minutes at constant temperature, in order to approach equilibrium slowly. The inclusions in olivine were studied by low-temperature microthermometry using Chaixmeca and Linkam TH-600 combined heating and cooling stages. The stage is calibrated in the temperature interval from -190° to 600°C using various natural and synthetic standards, as described by Roedder (1984); pure CO₂ in the inclusions in quartz was used for calibration at -56.6°C.

Characteristics of Melt Inclusions

Classification

All silicate minerals investigated in dunite and lherzolite contain fluid and/or melt inclusions; our main focus was on inclusions in olivine and pyroxene. It should be emphasized that the observations described here make the distinction between fluid inclusions and trapped melts (sulfide inclusions, crystallized polyphase inclusions). Several generations of fluid and melt inclusions may be texturally and petrographically recognized.

Sulfide inclusions (I). Sulfide inclusions in chromites and olivines (Figs. 2B and 2F) from dunite and sulfide-rich lherzolite occur as primary inclusions. The inclusions are angular and/or roughly spherical. They commonly occur parallel to growth planes of the host mineral (Fig. 2B). These inclusions generally range from 5 to 80 μ m in diameter and comprise several daughter minerals. Examination of the inclusions by reflection microscopy and electron microprobe analyses (EMPA) has shown that such inclusions consist of nickeliferous pyrrhotite, pentlandite, and chalcopyrite.

The sulfide melt nucleated on the surface of crystals growing from magmas and thus formed primary inclusions (Lorand and Ceuleneer, 1989). This suggests that the sulfide was enclosed as melt droplets during chromite crystal growth.

Crystallized polyphase inclusions (II). Such inclusions were found in all of the lherzolite and dunite samples that were studied. Most inclusions occurred within the interiors of olivine grains (Figs. 2C-2F) and are unrelated to fractures within the grains. The inclusions occur as single, elongate, isolated inclusions, sometimes coexisting with fluid inclusions. Individual inclusions may be 50 μ m, though more typically 15 to 35 μ m, in width. The inclusions often contain a vapor bubble (or shrinkage bubble) and daughter-crystal minerals at room temperature. The daughter crystals generally grow on the walls of the inclusions (Fig. 2E). Some inclusions

mimic the negative crystal shapes of their host minerals (Fig. 2D).

The fact that the inclusions occur as negative crystal shapes argues strongly for their origin as melt droplets rather than as solid aggregates at the time of their entrapment. The presence of daughter minerals indicates that the melt was slowly cooled after trapping (Roedder, 1984).

Fluid inclusions (III). Such inclusions were found in all of the lherzolite and dunite samples as primary and secondary inclusions. Primary fluid inclusions usually occur in olivine and pyroxene as spherical to tubular isolated inclusions and/or as "stratiform" ones along mineral growth zones. The size of the inclusions ranges from less than 10 μm to 25 μm ; they often occur as clusters of various shapes and sizes (Fig. 2G) and sometimes coexist with crystallized polyphase inclusions (Type II).

Secondary fluid inclusions form irregular grains in all of the silicate minerals. They often occur along short, healed fractures that extend to grain surfaces.

Type and composition of daughter minerals

Olivine, pyroxene, amphibole, and chrome-spinel daughter minerals were identified in host olivine and pyroxene by Laser Raman microspectrometry and EMPA (Table 1). Olivine is one of the major crystallized daughter minerals and often is present as aggregates of olivine microlite, growing on inclusion walls. The extinction of the daughter minerals does not correspond with that of the host minerals (Fig. 2C). Laser Raman analysis shows broad peaks at 821, 851, and 956 cm^{-1} .

Orthopyroxene accounts for 10% of the volume of inclusions and commonly is distributed at inclusion edges. It sometimes is located in the central part of inclusions, with parallel extension (Fig. 2D).

Chrome-spinel commonly occupies 5 to 20 vol% and is distributed in the central parts or along the walls of inclusions as opaque to

semiopaque, perfectly rounded or spherulitic aggregates (Fig. 2C). Their similar phase proportions and very elongate outlines suggest that they do not represent trapped mineral grains but rather are completely crystallized melts. The Laser Raman analyses show that these minerals are Cr-spinels, as indicated by the broad peaks at 572, 700, and 840 cm^{-1} (Fig. 3).

Laser Raman analyses of sulfide melt inclusions give peaks at 842, 916, and 1002 cm^{-1} (standard pyrite typically gives peaks at 842, 912, and 1004 cm^{-1}). Qualitative electron-microprobe analyses indicate that the sulfide melt inclusions are solid solutions of S-Fe-Ni. Sometimes, minor amounts of Cr and Cu were identified.

The sulfide inclusions in chromite typically are FeS with 0.0 to 0.44 wt% Ni and minor Co and Cu (Table 2). This finding is consistent with that of Skinner and Peck (1969), who reported that globules of immiscible sulfide melt in quenched samples from the Alae lava lake, Hawaii, also consist of homogeneous pyrrhotite. Analyses also revealed Ni-rich sulfide inclusions, containing 0.43 to 0.44 wt% Ni with Fe/Ni ratios in the range of 145.38 to 148.10 (atom%). Symmetrical compositional variations of Ni, Fe, and S (atom%) in sulfide inclusions from central to boundary areas of the chromite are shown in Figure 4. The data summarized in Table 2 show that sulfide melt inclusions are slightly iron rich and relatively sulfur and nickel poor at the core, and iron poor and nickel and sulfur rich along the rim of the chromites (Fig. 4).

Furthermore, it is interesting to note that the compositions of the sulfide inclusions differ slightly from that of intergranular pyrrhotite, which exhibits a distinctly narrow range of S compared to sulfide inclusions in chromite (Fig. 5). Taking the aforementioned facts into consideration, the paragenesis of the inclusions implies that the melts were $\text{CaO-MgO-Al}_2\text{O}_3\text{-Cr}_2\text{O}_3\text{-FeS}$ rich.

TABLE 1. Representative Chemical Compositions of Inclusion Daughter Minerals and Host Minerals (in wt%)

Sample no.:	1			2			3	
	h	d		h	d		h	d
	Ol	Ol	Opx	Opx	Ol	Opx	Cpx	Opx
SiO ₂	38.79	40.58	57.30	55.52	43.98	42.64	56.45	47.75
TiO ₂	0.06	0.10	0.07	0.15	0.07	0.06	0.04	0.06
Al ₂ O ₃	0.14	0.08	0.35	0.95	0.72	0.59	0.26	0.37
Cr ₂ O ₃	0.39	0.00	0.12	0.08	0.00	0.05	0.26	0.10
FeO	15.48	15.03	10.58	11.64	8.69	8.10	2.24	5.58
MnO	0.25	0.01	0.28	0.34	0.18	0.11	0.03	0.14
MgO	44.93	42.85	30.03	30.32	38.61	33.48	26.70	37.99
CaO	0.01	0.00	0.30	0.08	0.12	0.12	13.45	0.26
NiO	0.31	0.11	0.04	0.10	0.01	0.08	0.00	0.00
K ₂ O	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.11
Na ₂ O	0.06	0.00	0.02	0.00	0.03	0.00	0.08	0.05
Total	100.42	98.76	99.0	99.18	92.41 ¹	85.23 ¹	99.55	92.41 ¹

¹Altered.

Abbreviations: h = host mineral; d = daughter mineral; Ol = olivine; Opx = orthopyroxene; Cpx = clinopyroxene.

Chemical compositions of inclusion fluid

Most Type-II and Type-III inclusions (Figs. 2E and 2G) in the investigated samples contain abundant fluid phases, but we mainly studied fluid compositions in the primary inclusion from host olivine. The inclusions were checked under a microscope to ensure characteristic primary inclusions before Raman microprobe analysis. In most cases, each inclusion was scanned for CO₂, CO, SO₂, H₂S, and CH₄; in some of the inclusions, N₂ also was analyzed.

Laser Raman analyses indicate that CO₂ is a major ingredient in the inclusions (Table 3). Figure 3 shows a typical Raman spectrum recorded for an inclusion in host olivine. The results show that the fluids consist mainly of 70.6 to 71.6 mol% CO₂, 12.6 to 16.0 mol% SO₂, 1.1 to 6.7 mol% CH₄, and 5.7 to 15.7 mol% H₂S (Table 3). However, no obvious CO signal above background noise was shown by the Raman data (Fig. 3). Some of the Raman spectra revealed the presence of

minor amounts of Cl₂ and N₂. However, the results are considered ambiguous and will not be discussed further.

Trapping Temperature and Pressure of Inclusions

Trapping temperature

The homogenization temperatures obtained from Type-II inclusions in dunite and sulfide-rich lherzolite show a bimodal distribution (Fig. 6A); IIa inclusions give high homogenization temperatures, between 1000° and 1075°C, with a frequency maximum at 1000° to 1050°C (Fig. 6A). However, for some of the IIa inclusions that did contain sulfide daughter minerals, the homogenization temperatures occurred at >1075°C. This homogenization temperature is probably somewhat low compared to the true liquidus temperature, because the sulfide

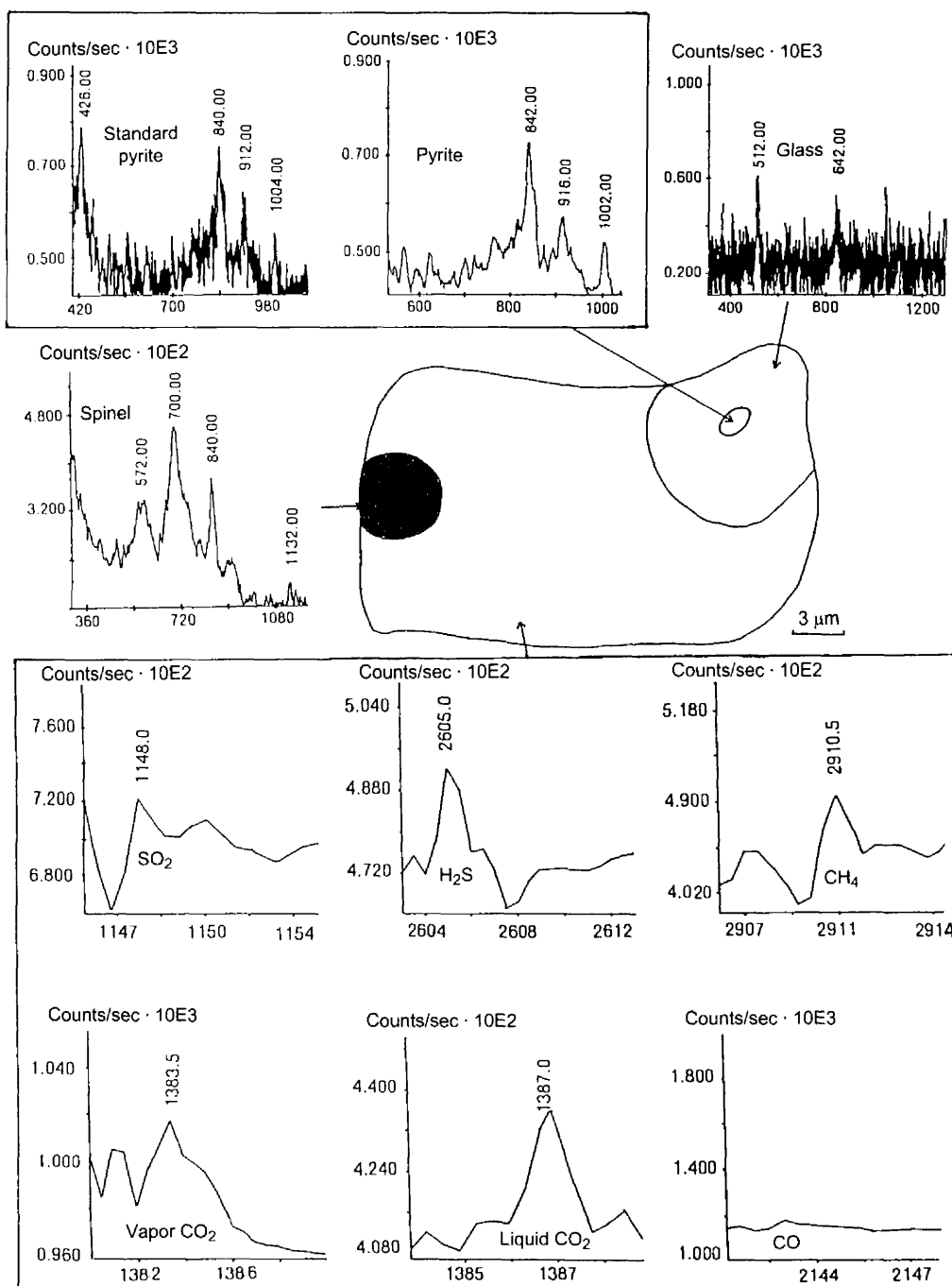


FIG. 3. Laser Raman spectra and compositions of inclusions in host olivine from the Jinchuan deposit ($\times 100$ magnification). Peaks at 1148.0 cm^{-1} , 2605.0 cm^{-1} , 2910.5 cm^{-1} , 1387.0 cm^{-1} , and 1383.5 cm^{-1} are for SO_2 , H_2S , CH_4 , and liquid and vapor CO_2 , respectively. Peaks at 572.0 cm^{-1} , 700.0 cm^{-1} , and 840 cm^{-1} are for spinel; at 512.0 cm^{-1} and 842.0 cm^{-1} for glass; and at 842.0 cm^{-1} , 916.0 cm^{-1} , and 1002.0 cm^{-1} for pyrite. Some small peaks have not been identified. The other characteristic frequencies of H_2 and O_2 were scanned, but no signal above background noise was detected. Raman shift is indicated on the X axis.

TABLE 2. Representative Chemical Compositions of Sulfide Inclusions in Chromite and Intergranular Pyrrhotite

Sample no.:	Sulfide inclusion					Intergranular pyrrhotite					
	1	2	3	4	5	6	7	8	9	10	11
	wt%										
Fe	64.27	63.13	63.22	62.95	62.22	60.60	60.87	61.81	61.36	61.92	61.35
Ni	0.00	0.00	0.01	0.02	0.06	0.43	0.44	0.48	0.37	0.04	0.29
Co	0.06	0.07	0.06	0.06	0.06	0.06	0.08	0.07	0.07	0.07	0.07
Cu	0.02	0.03	0.04	0.04	0.05	0.01	0.04	0.04	0.00	0.03	0.00
S	36.07	37.11	36.87	37.26	38.05	39.40	38.95	38.06	38.64	37.37	38.78
Total	100.42	100.34	100.20	100.33	100.44	100.50	100.38	100.46	100.44	99.43	100.49
	atom%										
Fe	50.54	49.38	49.57	49.20	48.37	46.73	47.10	48.05	47.54	48.71	47.47
Ni	0.00	0.00	0.01	0.01	0.04	0.32	0.32	0.36	0.27	0.03	0.21
Co	0.04	0.05	0.04	0.04	0.04	0.04	0.06	0.05	0.05	0.05	0.05
Cu	0.01	0.02	0.03	0.03	0.03	0.01	0.03	0.03	0.00	0.02	0.00
S	49.40	50.55	50.35	50.71	51.51	52.91	52.49	51.52	52.14	51.19	52.26
Fe/Ni			6643.48	3307.55	1089.73	148.10	145.38	135.32	174.27	1626.72	222.31
Fe/S	1.02	0.98	0.98	0.97	0.94	0.88	0.90	0.93	0.91	0.95	0.91

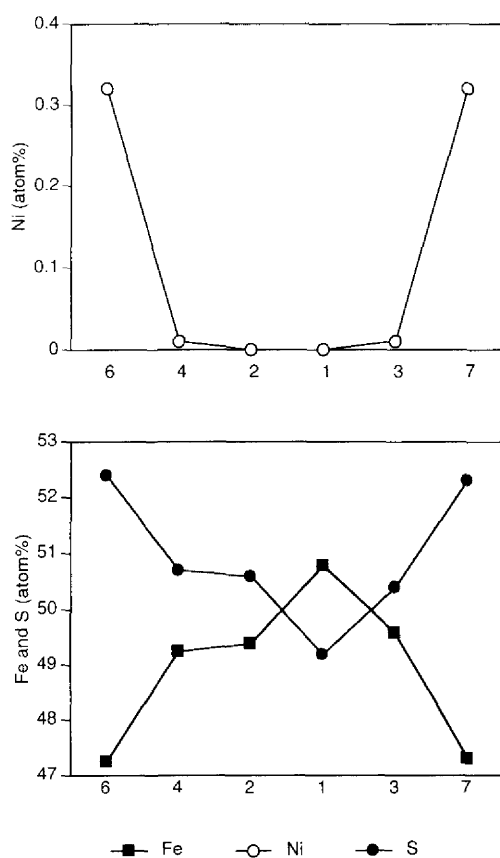


FIG. 4. Variation diagram of Ni, Fe, and S (atom%), showing that the chemical composition of primary sulfide inclusions along the rim of chromite is characterized by a considerable increase in Ni and S, accompanied by a decrease in Fe. Analyses were made by EMPA in Figure 2B and correspond to those in Table 2.

daughter minerals often failed to dissolve completely in the melt, even at temperatures 20° to 40°C above the melting point of host olivine along the inclusion walls. Because of the very small volumes involved, such inclusions were still considered to have approached their liquidus temperatures closely (Hansteen, 1991).

Some inclusions contain dark brown to semi-transparent spinel grains. The spinel often failed to dissolve completely in the melt, even at temperatures 50°C above final olivine melting. Hansteen (1991) and Han-

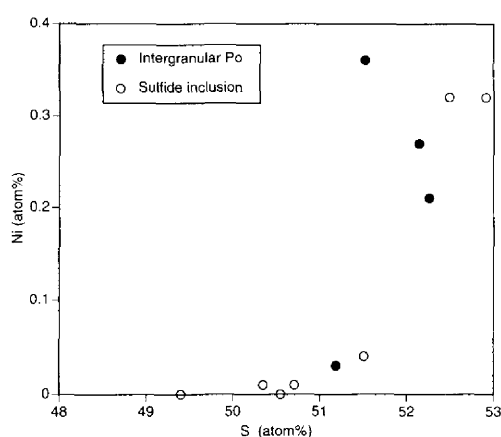


FIG. 5. Variation diagram of Ni versus S in sulfide inclusions and in intergranular pyrrhotite occurring outside chromite grains (data from Table 2).

steen et al. (1991) also observed this phenomenon when they considered they had approached the liquidus temperatures closely.

Heating experiments also were performed on Type-II inclusions in the sulfide-poor lherzolite. The results indicate that the majority of the inclusions nearly homogenized into silicate melt at the minimum temperature in the range of 900° to 975°C. The data display a maximum at 900° to 950°C (Fig. 6B). This homogenization temperature corresponds to the temperature ranges of Type-IIb inclusions in sulfide-rich lherzolite and dunite (Fig. 6A).

Type-III inclusions contained daughter minerals that homogenized into the silicate melt in the temperature interval from 900° to 950°C and showed incipient partial-melting reactions at 750°C. These homogenization temperatures probably are somewhat low compared to the true liquidus temperature because large amounts of volatiles existed in the inclusions. The largest recorded homogenization temperature of 950°C is therefore regarded as a minimum value of liquidus temperature for the trapped silicate melts.

TABLE 3. Representative Laser Raman Analyses of Inclusions in Olivine

Sample no.:	1		2	
	Peaks, cm ⁻¹	Contents, mol%	Peaks, cm ⁻¹	Contents, mol%
CO ₂	1383.5-1387.0	71.6	1388.0	70.6
SO ₂	1148.0	16.0	1150.5	12.6
CH ₄	2910.5	6.7	2915.5	1.1
H ₂ S	2605.0	5.7	2611.5	15.7

Trapping pressure

Trapping pressures were estimated from molar volumes of primary pure CO₂ and CO₂-dominated fluid inclusions in olivine from Iherzolite and dunite. Examination of 20 pure CO₂ inclusions and CO₂-dominated fluid inclusions indicated that samples of sulfide-poor Iherzolite define a Th (liquid-vapor homogenization temperature) in the range of +22° to +31°C, with a frequency maximum at +26° to +29°C (into liquid phase, critical point of CO₂ at 31.1°C) (Angus et al., 1976). This corresponds to a molar volume of 63 to 71 cm³/mole.

Sixteen pure CO₂ inclusions and CO₂-dominated inclusions in sulfide-rich Iherzolite and dunite samples yielded homogenization temperatures from +22° to +31°C, with a maximum at +26° to +29°C, corresponding to molar volumes of the inclusions from the sulfide-poor Iherzolite. Eleven inclusions displayed homogenization temperatures from +10° to +15°C, with a marked peak at +12.4°C (into liquid phase), corresponding to a molar volume of 51 to 54 cm³/mole.

Discussion

Origin of melt inclusions

Many geologists believe that immiscibility does exist in petrogenic and metallogenic processes for Cu-Ni-bearing mafic-ultramafic intrusions (Skinner and Peck, 1969; Naldrett, 1981; Mountain et al., 1987). Some valuable ore deposits have formed by

the process of separation of this immiscible sulfide melt. Therefore it is important to know at which stage of a differentiation sequence the sulfur became saturated with respect to silicate melt. Fortunately, immiscible sulfide melt inclusions provide this information.

The distribution of the sulfide melt inclusions (Type I, Fig. 2B) suggests that the sulfide was trapped as liquid droplets during chromite crystallization. Sulfide inclusions trapped in the core of chromite grains imply entrapment at a relatively early stage and indicate that immiscible sulfide melt separated prior to, or simultaneously with, chromite crystallization. Andersen et al. (1987) indicated that the sulfide inclusions in chromite strongly suggest entrapment from an immiscible sulfide liquid during chromite crystallization. Cameron (1982) reached a similar conclusion after studying the Merensky Reef in the Bushveld Complex.

Sulfide melt inclusions parallel to the growth surfaces of olivines indicate that sulfide melt was enclosed as melt droplets during olivine crystal growth. They also suggest that sulfide melt segregation occurred prior to, or simultaneously with, olivine crystallization.

Rounded chrome-spinel is commonly discovered in Type-IIa inclusions from olivine (Fig. 6). This phenomenon may be explained in two ways: (1) entrapment of early Cr-spinel and chromite crystals; and (2) the crystallized production of trapped melt. Their similar phase proportions and very rounded

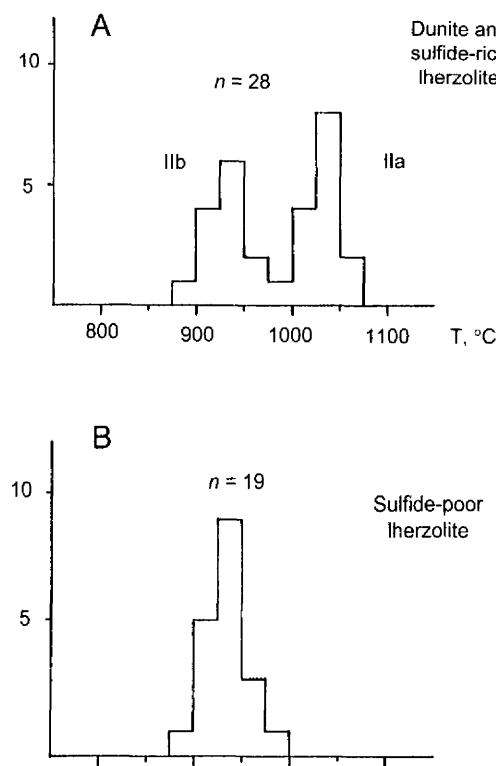


FIG. 6. Histogram of homogenization temperatures of Type-II inclusions in olivine from dunite and sulfide-rich lherzolite (A) and sulfide-poor lherzolite (B). Abbreviations: Ila = petrographically early inclusions; Ilb = petrographically late inclusions. See text for further explanation.

shapes suggest that rounded chrome-spinel does not represent trapped mineral grains but crystallized melt inclusions.

P-T history

Pressure-temperature conditions can be estimated from the fluid and melt inclusions. Fluid inclusions in olivine grains from lherzolite and dunite have been used to estimate pressure conditions for magmatic processes in the Jinchuan intrusion. It is of interest that the homogenization temperatures have quite distinctive characteristics.

1. The homogenization temperatures of +10° to +15°C, corresponding to minimum molar volumes of 51 to 54 cm³/mole, indicate a filling pressure of ~4.3 to 5 kbar at

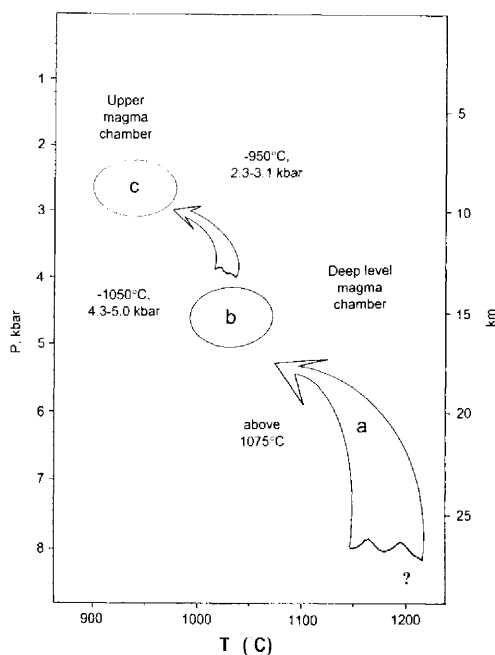


FIG. 7. P-T evolution as interpreted from fluid and melt inclusions in sulfide-poor lherzolite, sulfide-rich lherzolite, and dunite. The deep chamber (b) corresponds to lower Th (+10° to +15°C) from dunite and sulfide-rich lherzolite. The upper chamber (c) corresponds to higher Th (+26° to +29°C) in primary inclusions from sulfide-poor lherzolite and in inclusions from dunite and sulfide-rich lherzolite. Arrows indicate ascending magma. Elliptical areas represent magma chambers. See text for further discussion.

1050°C, on the basis of Bottinga and Richet's (1981) equation for the CO₂ state. This is equivalent to entrapment at depths of 12.9 to 15 km, assuming lithostatic pressure and an average crustal density of 3.0 g/cm³. Inclusions of such densities are found in olivine from dunite and sulfide-rich lherzolite, suggesting that at least some of the samples originated at this depth, which may represent a deep-level magma chamber (b in Fig. 7).

2. The low-density homogenization peak (+26° to 29°C) of primary fluid inclusions in olivine from the sulfide-poor lherzolite indicates a molar volume of 63 to 71 cm³/mole, corresponding to a filling pressure of ~2.3 to 3.1 kbar at 950°C and a depth of ~6.9 to 9.3 km (the lithostatic value of 3.0 g/cm³ is adopted). Inclusions of such densities are

found in olivine from sulfide-poor lherzolite, indicating that at least some of the samples originated at depths from 6 to 9 km, which may represent the depth of an upper magma chamber (c in Fig. 7).

On the other hand, amphibole was found in the intrusion, which suggests a pressure >2 kbar (Spulber and Rutherford, 1983). If 2 kbar represents a minimum pressure at the top of the intrusion, then the pressure was at least 3.1 kbar at the base of the intrusion. A sequence of up to 6.5–10 km of supracrustal sediments and lavas around the intrusion is in agreement with estimates from the inclusions (SGT, 1984). From these considerations, we conclude that 2 to 5 kbar is a plausible estimate of the pressure range during crystallization of the dunite and lherzolite.

Temperature evolution can be interpreted from polycrystalline fluid inclusions. A liquidus temperature of 1075°C suggested by Type-II inclusions from dunite and sulfide-rich lherzolite is regarded as the minimum value of the magmatic system (a in Fig. 7). It agrees well with the experimentally established liquidus temperature of 1100°C at 1 atm for natural basalt (Takahashi, 1980).

A solidus temperature of ~900°C is indicated by Type-II inclusions from sulfide-poor lherzolite. This result agrees well with the solidus temperatures indicated by Type-III inclusions containing daughter minerals. The temperatures of these inclusions therefore also represent the formation temperatures of dunite and lherzolites.

These relationships suggest that the parental magma intruded the deep magma chamber at a depth of 12.9 to 15 km. Temperature and pressure reductions produced sulfide melt immiscibility and an early-stage separation of the sulfide melt in the deep magma chamber (b in Fig. 7). Gravity fractionation could result in the accumulation of sulfide melt at the base of the magma chamber and the crystallization of olivine in some dunites and sulfide-rich lherzolites. This may have been followed by intrusion into the upper magma chamber at a depth of 6.9 to 9.3 km

(c in Fig. 7) and formation of the sulfide-poor lherzolite.

In the deep magma chamber, the solubility of sulfides might be decreased in the melt, owing to an abrupt fall in total pressure and temperature (Haughton et al., 1974; Roeder, 1979, 1981; Mountain et al., 1987). Decrease of sulfur solubility can result in sulfur saturation and segregation of a sulfide melt. Because of their high liquidus temperature, the parental magmas probably had a considerable capacity to dissolve sulfur and likely were sulfide undersaturated before sulfide melt segregation.

Deformation textures are common in dunite and sulfide-rich lherzolite. Various deformation textures have been explained by magma-chamber processes (e.g., Coleman, 1977; Nicolas, 1989). Thus deformation textures in the dunite and sulfide-rich lherzolite are similarly ascribed to deep-magma-chamber processes. This result is consistent with this study and indirectly supports the authors' ideas.

Compositional evolution of sulfide melts

Variation in the composition of individual sulfide melt inclusions along different growth surfaces of chromite grains, from core to rim (Fig. 4), is ascribed to enclosure of immiscible sulfide melt at various stages of its evolution. It is of interest that the compositions of the primary sulfide melt inclusions enclosed in chromite grains show symmetrical compositional variations. The Ni concentrations of sulfide melt inclusions exhibit a gradual increase outward from the central part of the chromite, which can be explained by slow fractionation processes of Ni. Hence, the increase of the Ni of sulfide melt inclusions toward the margin in chromite reflects gradual enrichment in Ni during fractionation as a result of incompatibility of Fe and relative compatibility of Ni and S in magma. It also agrees well with experimental results for the Cu-Ni-Fe-S system (Craig and Kullerud, 1969). The peculiar and varied composition of the sulfide inclusions suggests that the

melt gradually changed in composition, depending on temperature and equilibration reactions, until the crystallization of sulfide had been completed.

Microprobe traverses across the inclusions reveal the presence of chemical gradients. The increase in Ni concentration is indicated by a steep Ni curve and an associated decrease in the Fe concentration (Fig. 4). This led us to conclude that certain sudden events might also result in the separation and enrichment of Ni in the melt.

Conclusions

Polyphase inclusions (the primary sulfide melt inclusion in chromite) and fluid inclusions in olivine are interpreted to represent droplets trapped during magma differentiation. Sulfide inclusions trapped in the core of chromite imply that the immiscible sulfide melt separated at a relatively early stage. The concentration of Ni in the sulfide melt was gradually enriched from the early to the late stage during the melt fractionation. The primary inclusion paragenesis indicates that the Jinchuan melts were $\text{CaO-MgO-Al}_2\text{O}_3\text{-Cr}_2\text{O}_3\text{-FeS}$ rich.

Laser Raman analyses show that the melt contained abundant volatiles, mainly CO_2 , but also small amounts of SO_2 , CH_4 , and H_2S . The results of thermometry suggest that at least some of the dunite and sulfide-rich lherzolite was formed in the temperature range of 1000° to 1050°C in a magma chamber at 12.9 to 15 km depth, whereas sulfide-poor lherzolite in the range of 900° to 950°C formed in the upper magma chamber at a depth of 6.9 to 9.3 km.

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